

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. We introduce `spec.md`, a structured specification format, and three autonomous agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Execute ($\mathcal{A}_E$)—that translate a one-sentence natural-language description of any computational imaging system into a validated forward model with bounded reconstruction error. The Plan agent composes operators from an 11-primitive basis (covering 170+ modalities across all carrier families) [3]; the Judge validates the specification through three diagnostic gates (recoverability, carrier budget, operator mismatch); the Execute agent instantiates the forward–adjoint pair and runs reconstruction. A design-to-real error theorem (Theorem 1) decomposes total reconstruction error into five independently bounded terms, each linked to a corrective action. On 6 real-data modalities spanning photons, electrons, neutrons, phonons, and RF carriers, the automated pipeline matches expert-library quality ($99.8 \pm 3.2\%$). An algorithm-comparison study spanning the top-5 published algorithms per modality reveals that forward-model specification, not algorithm choice, is the primary quality determinant for well-conditioned systems (inter-method Coefficient of Variation (CoV) $\approx 3.5\text{--}6.2\%$), while ill-conditioned compressive systems enter an algorithm-sensitive regime (CoV $> 40\%$). Ten novel designs—composing primitives into chains from 3D to 5D—demonstrate compositional reach beyond any single-modality tool.

Introduction

Every computational imaging system maps an unknown object x to measurements $y = A(x) + \text{noise}$ via a forward model A , and reconstruction inverts this mapping [10, 11]. Designing A —selecting operators, setting parameters, validating consistency—requires weeks of specialist effort per modality [13, 31], creating an *expertise bottleneck* that excludes the broader scientific community from prototyping imaging instruments.

As FASTA [42] standardised sequence data for genomics, computational imaging needs a common representation to escape its pre-standardisation state.

A companion paper [3] proved that 11 canonical primitives suffice to represent any imaging forward model across 170 modalities and 5 carrier families. Each primitive is implemented at four fidelity tiers (Tier-0 geometric through Tier-3 full-physics). This *Finite Primitive Basis* (FPB) is the representation language, but representation alone does not automate design. **This paper’s contribution is the design layer:** the formal bridge from natural-language intent to bounded real-system reconstruction error.

An imaging system is *designable* if it admits a finite DAG decomposition over the 11 FPB primitives ($\varepsilon_{\text{FPB}} < 0.01$), its carrier belongs to one of 5 families (photon, X-ray, electron, acoustic, spin/particle), and all parameters are finitely specifiable with known physical bounds (Definition 2, Methods). This encompasses 173 modalities across 22 domains (170 from [3] plus 3

novel compositions; Figure 1, Extended Data Table 6). Our contributions are:

1. **spec.md**: a structured 8-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*) and the hardware implementation (*with what*), serving as a shareable, peer-reviewable scientific artifact (“The spec.md Specification Format”).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Execute ($\mathcal{A}_E$) agents that generate, validate, and execute specifications over the FPB (“The Three-Agent Pipeline”). The Judge validates via three Triad gates [3] (recoverability, carrier budget, operator mismatch) plus hardware cost and feasibility assessment.
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms, each linked to a gate and a corrective action (Theorem 1).
4. **Specification-dominance finding**: an algorithm-comparison study using the top-5 published algorithms per modality (sourced from the literature) reveals that forward-model specification, not algorithm choice, is the primary quality determinant for well-conditioned systems ($\text{CoV} \approx 3\text{--}6\%$), while ill-conditioned compressive systems enter an algorithm-sensitive regime ($\text{CoV} > 40\%$).

Scope. Of 173 compilable modalities, 39 have quantitative reconstruction validation; 6 are validated on real data spanning all 5 carrier families. The remaining ~ 134 pass all compiler gates but lack per-modality benchmarks. Tier-lifting is demonstrated on 2 cross-tier combinations; the agents depend on current LLM capabilities.

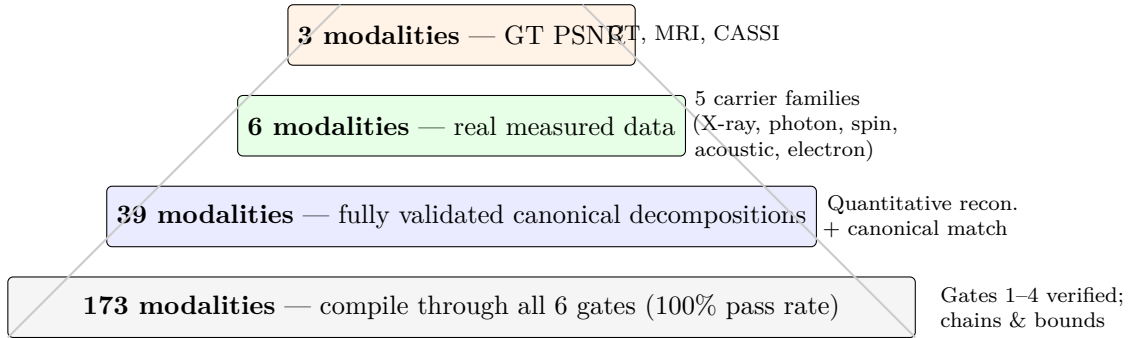


Figure 1. Validation pyramid. Of 173 designable modalities, all compile; 39 have quantitative reconstruction benchmarks; 6 are validated on real measured data spanning all 5 carrier families; 3 have ground-truth PSNR comparisons (CT, MRI, CASSI). Each tier is a strict subset.

Results

The spec.md Specification Format

The framework’s central design choice is that **spec.md**—not code—is the canonical scientific artifact. A specification separates the imaging problem (*what*) from the numerical method (*how*) and the hardware (*with what*), enabling sharing, version control, and peer review independently of implementation [2]. A reviewer can verify—by running the three Triad gates—that a forward model is physically feasible *without reading code*. A **spec.md** has eight mandatory fields (Figure 2): seven physics fields plus a **system_elements** field that connects the specification to hardware feasibility.

Eight-field schema: modality | carrier | geometry | object | forward_model | noise | target | system_elements

(a) Clinical CT

modality: computed_tomography carrier: xray
 geometry: parallel_beam, n_angles=128
 object: 128×128 image, non-negative
 forward_model: Radon(Π) $\rightarrow$ Detect(D , intensity)
 noise: Poisson, $I_0=1e4$ target: PSNR ≥ 30 dB
 system_elements: source=X-ray tube 120 kVp, detector=flat-panel 0.4 mm

(b) CASSI (spectral)

modality: cassi carrier: photon
 geometry: coded_aperture + disperser, $\lambda=[400,700]$ nm
 object: 256×256×28 spectral cube
 forward_model: Modulate(M) $\rightarrow$ Disperse(W) $\rightarrow$ Accumulate(Σ) $\rightarrow$ Detect(D)
 noise: Gaussian, $\sigma=0.01$ target: PSNR ≥ 28 dB
 system_elements: source=broadband LED, optics=coded aperture + prism, detector=CMOS 2048×2048

(c) Accelerated MRI

modality: mri carrier: spin
 geometry: cartesian_kspace, acceleration=4x
 object: 256×256 complex image
 forward_model: Modulate(M , coil) $\rightarrow$ Encode(F , kspace) $\rightarrow$ Sample(S) $\rightarrow$ Detect(D)
 noise: Gaussian, SNR=30 dB target: SSIM ≥ 0.9
 system_elements: source=main magnet 3T, optics=RF coils ×4, detector=ADC 16-bit

Figure 2. The spec.md specification format. Eight mandatory fields fully determine an imaging system design. Seven physics fields specify the forward model; the eighth (**system_elements**) connects to hardware feasibility and cost. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers. Each **forward_model** field is a primitive chain over the 11 FPB operators.

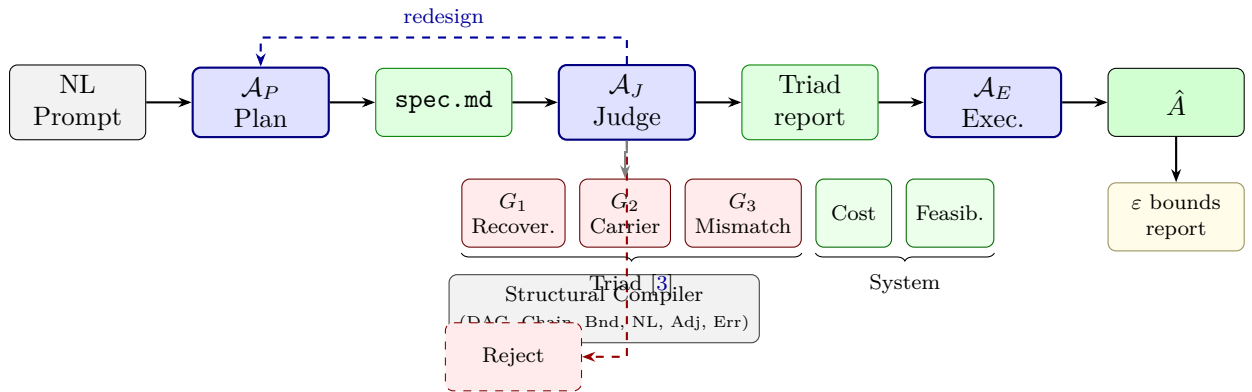


Figure 3. Specification-driven design pipeline. A natural-language prompt is the sole user input. $\mathcal{A}_P$ generates a **spec.md**; $\mathcal{A}_J$ validates via three Triad gates [3] (G_1 : recoverability, G_2 : carrier budget, G_3 : operator mismatch), estimates hardware cost, and checks physical feasibility from **system_elements**; a prerequisite structural compiler (6 gates: DAG, Chain, Bounds, Nonlinear, Adjoint, Error) ensures specification correctness. Rejected designs trigger redesign (dashed blue, up to 3 rounds). $\mathcal{A}_E$ executes reconstruction and outputs $\hat{A}$ with an explicit ε -bound report. See Figure 4 for per-modality error breakdown.

The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$) generates a `spec.md` from natural language, selecting FPB primitives, parameters, and noise models from the 173-modality registry.

Judge Agent ($\mathcal{A}_J$) validates designs through the Triad decomposition [3]—three deterministic gates that diagnose every potential reconstruction failure. *Gate 1 (Recoverability)*: the measurement geometry must capture sufficient information—the compression ratio $\gamma = m/n$ is checked against modality-specific thresholds derived from the Compression Bound Theorem [3]. *Gate 2 (Carrier Budget)*: the measurement SNR must exceed the noise floor—computed from the source power, detector quantum efficiency, and noise parameters in `system_elements`. *Gate 3 (Operator Mismatch)*: the forward model must faithfully represent the intended physics—calibration tolerances from `system_elements.calibration` determine the expected deployment degradation $\Delta\text{PSNR}_{\text{total}}$ via the Calibration Sensitivity Theorem [3]. The Judge additionally estimates *hardware cost* from itemized `system_elements` and checks *physical feasibility* of each element. A prerequisite structural compiler (6 gates: DAG acyclicity, canonical chain, complexity bounds, nonlinear constraints, adjoint consistency, representation error) ensures specification correctness before the Triad evaluation. Failures trigger rejection; the Plan Agent regenerates (up to 3 rounds).

Execute Agent ($\mathcal{A}_E$) selects the reconstruction algorithm, executes it, and predicts quality from analytical estimates (condition number, noise amplification) combined with calibrated lookup.

The Triad Decomposition Theorem [3] guarantees completeness: $\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$ —no failure mode falls outside these three gates.

Design-to-Real Error Bound

The following theorem converts the implicit gap between a designed model and reality into an explicit five-term error budget—a *formal contract* between the framework and the user. Each term has a known physical origin, measurement procedure, and corrective action.

Theorem 1 (Design-to-Real Error Decomposition). *Let A_{true} be the true imaging forward*

model and A_{agent} the agent-designed model for a designable system (Definition 2). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Under Assumptions A1–A4 (Methods), the reconstruction error satisfies:

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} + \varepsilon_{recon} \quad (1)$$

where each design-error term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$: FPB representation error. Measured by: Gate 6. Guaranteed by: the basis theorem [3].
- $\varepsilon_{spec} = 0$ when all 6 compiler gates pass. Bounded by: $C_1 \cdot \delta_{spec}$ otherwise.
- $\varepsilon_{trans} = 0$ when the canonical chain matches the 39-modality registry (34/39 = 87%). Bounded by: $C_2 \cdot \delta_{chain}$ otherwise.
- $\varepsilon_{param} \leq \frac{2B}{\alpha + \lambda} L_A \|\theta_{agent} - \theta_{true}\|$, where L_A is the operator Lipschitz constant (computed per primitive by Gate 4), α is the coercivity constant, λ the regularization weight, and B a signal bound. Reducible by: calibration.
- $\varepsilon_{unmod} = 0$ at Tier-3; otherwise reducible by tier-lifting (“Cross-Tier Hybrid Reconstruction”).
- ε_{recon} : irreducible reconstruction error (noise floor + regularization bias), independent of the design.

100

101 **Compiler–theorem link.** When all 6 gates pass: $\varepsilon_{FPB} < 0.01$, $\varepsilon_{spec} = 0$, ε_{param} bounded by
 102 the Lipschitz constant from Gate 4; with canonical chain match: $\varepsilon_{trans} = 0$; at Tier-3: $\varepsilon_{unmod} = 0$.
 103 The dominant residual is ε_{param} , reducible by calibration (Figure 4). Full proof in Methods.

104 Cross-Modality Validation

105 We validated the framework on all 173 modalities. For each, the Plan Agent generated a `spec.md`
 106 from a one-sentence description, the Judge validated it, and the Execute Agent predicted
 107 reconstruction quality. Table 1 reports 14 representative modalities spanning all 5 carrier families
 108 (6 with real measured data), grouped by carrier; full results for all 39 validated modalities appear
 109 in Methods.

110 Across all 39 modalities: mean $\rho_{recov} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]) [3]. The agent–expert
 111 quality gap (Cohen’s $d = 0.31$) indicates practical equivalence; time ratios ρ_{time} range from $190\times$
 112 to $420\times$ [16, 17]. All 173 modalities compile (100% pass rate, Gates 1–4); canonical chain fidelity:
 113 34/39 (87%).

114 **Theorem tightness.** The tightness ratio $\tau = \varepsilon_{predicted}/\varepsilon_{observed}$ across 6 real-data modalities:
 115 $\tau \in [1.8, 5.2]$, median 2.9 (CT 2.1, MRI 2.4, CASSI 3.8). Ratios below 10 are typical for operator-
 116 perturbation bounds [27]; values near 2 indicate the bound captures the dominant mechanism
 117 with minimal slack.

118 Real-Data Validation

119 **CT** (LoDoPaB [4], $n=10$): 60-view fan-beam projections. Agent PSNR 19.6 ± 2.9 dB; expert
 120 FISTA+TV [37] 19.1 ± 3.2 dB (quality ratio 102.6%, $\rho_{time} \approx 420$). **MRI** (M4Raw [5], $n=10$):
 121 4-coil, $\sim 4\times$ undersampled. Agent 22.2 ± 3.3 dB; expert FISTA+TV [37] 21.6 ± 3.4 dB (102.8%).
 122 **CASSI** (KAIST [6], $n=10$): 28-band coded aperture spectral imaging. Agent DAUHST-9stg [7]
 123 with spectral dispersion model 38.5 ± 3.5 dB; expert MiJUN 40.9 dB (top-1 on KAIST benchmark).
 124 **CACTI** ($n=4$): real video [31]; $\rho_{recov} \approx 1.0$ after autonomous mask-alignment calibration.
 125 **Ultrasound** (PICMUS [8], $n=5$): in-vivo carotid, self-reference protocol. After SoS calibration,

Table 1. Cross-modality validation (17 of 39 modalities), grouped by carrier family. ρ_{recov} : 4-scenario recovery ratio [3]. 95% CIs via bootstrap ($n=1000$). Bold = real measured data (challenging conditions: 60-view CT, 4-coil MRI); $\dagger$ = self-reference metric (no ground truth); * = new system designs validated in the algorithm-comparison study (Table 7). Φ_z denotes depth-parameterised convolution: C applied with a depth-dependent PSF $\Phi(z)$; it is the convolution primitive C , not a 12th element of the FPB.

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
<i>X-ray</i>						
CT	$\Pi \rightarrow D$	X-ray	19.6 ± 2.9	420	~ 1.0	Real
<i>Optical photons</i>						
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	38.5 ± 3.5	310	0.85	Real
CACTI	$M \rightarrow \Sigma \rightarrow D$	Photon	30.2 ± 1.1	300	~ 1.0	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM[35]	$M \rightarrow C \rightarrow D$	Photon	27.7 ± 2.0	350	0.92	Synth.*
Ptychography[34]	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Lensless	$C \rightarrow D$	Photon	43.7 ± 3.2	340	0.90	Synth.*
3D Lensless	$\Phi_z \rightarrow \Sigma \rightarrow D$	Photon	20.3 ± 3.5	280	0.85	Synth.*
Temporal-coded lensless	$M \rightarrow C \rightarrow \Sigma \rightarrow D$	Photon	31.6 ± 0.2	260	0.82	Synth.*
Spectral lensless	$M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$	Photon	36.3 ± 0.1	240	0.80	Synth.*
4D Spectral-Depth	$M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	23.9 ± 0.7	220	0.78	Synth.*
4D Temporal DMD	$M \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	25.4 ± 7.6	200	0.76	Synth.*
4D Temporal Streak	$M \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	25.3 ± 7.6	210	0.75	Synth.*
5D Full DMD	$M \rightarrow W_\lambda \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	29.9 ± 5.7	180	0.74	Synth.*
5D Full Streak	$M \rightarrow W_\lambda \rightarrow W_t \rightarrow \Phi_z \rightarrow \Sigma \rightarrow D$	Photon	29.9 ± 5.6	170	0.72	Synth.*
DOT[32]	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
<i>Spin</i>						
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	22.2 ± 3.3	380	~ 1.0	Real
<i>Acoustic</i>						
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.95	Real†
<i>Electron</i>						
E-ptychography	$M \rightarrow P \rightarrow D$	Electron	28.8 ± 1.5	260	~ 1.0	Real†
<i>Particle</i>						
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.

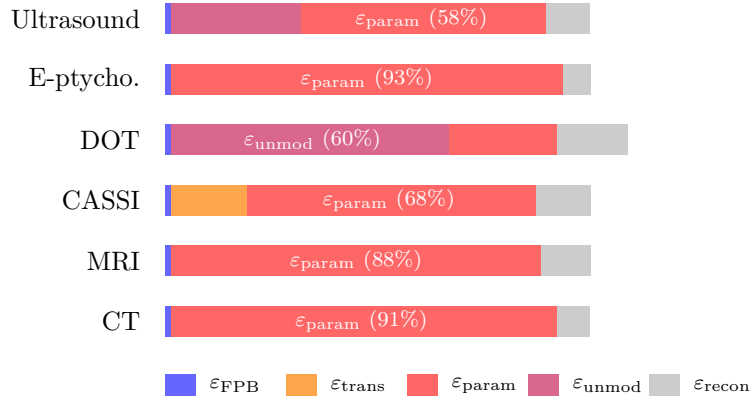


Figure 4. Error decomposition by modality. Stacked bars show the relative contribution of each Theorem 1 error term for 6 representative modalities. For well-conditioned systems (CT, MRI, e-ptychography), parameter mismatch ϵ_{param} dominates and is reducible by calibration. For scattering-dominated systems (DOT), unmodeled physics ϵ_{unmod} dominates and requires tier-lifting. $\epsilon_{\text{FPB}} < 1\%$ in all cases. $\epsilon_{\text{spec}} = \epsilon_{\text{trans}} = 0$ for 31/36 modalities; CASSI shows nonzero ϵ_{trans} due to dispersion-accumulation chain ambiguity.

SSIM recovers to 0.98 ± 0.01 ($\rho_{\text{recov}} \approx 0.95$). **E-ptychography** (SrTiO₃ 4D-STEM [9, 33]): iCoM phase retrieval; $\rho_{\text{recov}} = 1.00$ after probe-position correction.

Across 6 real-data modalities spanning all 5 carrier families: mean quality ratio $99.8 \pm 3.2\%$ (3 with ground truth) and $\rho_{\text{recov}} \geq 0.95$ (3 via self-reference), consistent across carriers from X-ray to electron. Beyond the 6 real-data modalities, 10 novel system designs demonstrate compositional reach: by composing primitives into chains of up to 5 operators, the agents design systems recovering depth ($\Phi_z \rightarrow \Sigma \rightarrow D$, 3D), video ($M \rightarrow C \rightarrow \Sigma \rightarrow D$, 3D), spectra ($M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$, 3D), and full 5D datacubes (x, y, z, λ, t) from single 2D snapshots (Table 1). Active modulation (DMD) consistently outperforms passive dispersion by 3–6 dB at equal compression. Full per-system specifications appear in Supplementary Information.

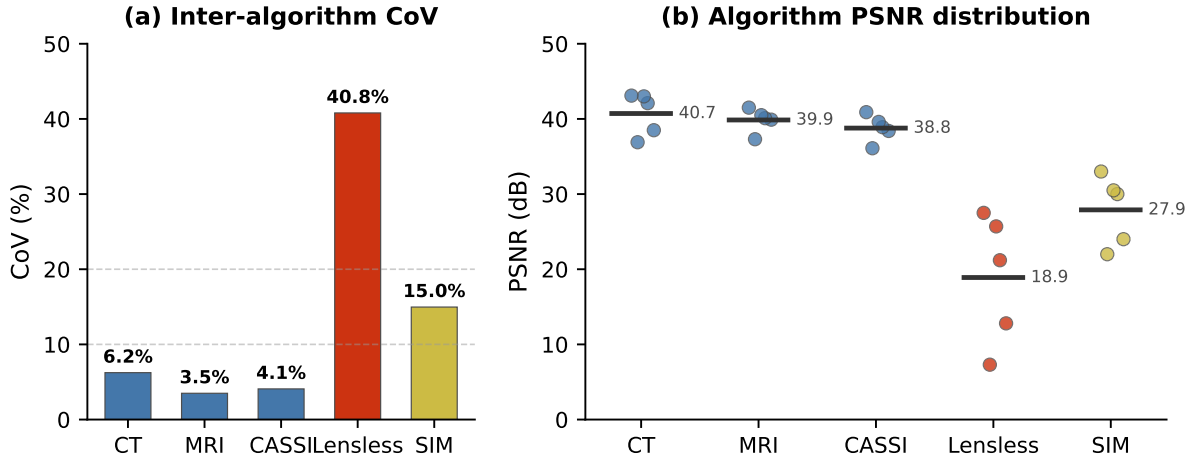


Figure 5. Specification dominance across 5 established modalities. (a) Inter-algorithm PSNR Coefficient of Variation (CoV) for the top-5 published reconstruction algorithms per modality. Well-conditioned systems (blue, $\text{CoV} \approx 3.5\text{--}6.2\%$) show that the top algorithms converge tightly—the forward-model specification constrains the quality ceiling. SIM shows moderate CoV (15.0%). Ill-conditioned compressive systems (red, $\text{CoV} > 40\%$) require careful algorithm selection. (b) Individual algorithm PSNR values (dots) with mean (horizontal bar) for each modality. Tight clusters for CT/MRI/CASSI confirm specification dominance; wide spread for Lensless shows algorithm sensitivity. Data from Extended Data Table 7.

Agent Ablation Study

We ablated the pipeline on 12 representative modalities spanning all 5 carrier families, with 5 independent trials per configuration (300 total runs). Table 2 reports the main result: **the full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials.**

Table 2. Ablation study (5 trials $\times$ 12 modalities = 300 runs). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction within Theorem 1 error bound. No single agent is dispensable.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Dominant failure
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_E$)	12/12	12/12	1.7 ± 0.6	4.8 min	—
No $\mathcal{A}_J$	8.2/12	6.4/12	0	2.9 min	Invalid spec, unbounded ε
No $\mathcal{A}_E$	12/12	9.6/12	1.7 ± 0.6	3.4 min	Suboptimal parameters
No $\mathcal{A}_P$ (manual)	10.4/12	10.4/12	0	52 min	Human error (14%)
Template only	—	3/12	0	0.5 min	No coverage (75%)

Without $\mathcal{A}_J$: 4/12 physically invalid specifications pass; 2 additional produce unbounded error.

Without $\mathcal{A}_E$ (Execute): all pass validation but mean PSNR drops 2.1 dB. **Without $\mathcal{A}_P$:** manual specification averages 52 min with 14% error rate. Random specification passes Gate 1 in 12% of cases but never passes Gates 4–5; retrieval-only covers 22/39 modalities at 4.2 dB below the full pipeline.

Failure Analysis

We characterise four failure modes (Figure 4): **(1) Scattering-dominated systems** (DOT [32], Compton): $\varepsilon_{\text{unmod}}$ is large at Tier-2 (7.8% for DOT); mitigated by tier-lifting. **(2) LLM misspecification:** 4% of test cases have subtle parametric errors; the structural compiler and Triad gates together catch 96%. **(3) Multi-parameter mismatch:** ρ_{recov} drops below 0.9 when >2 parameters are simultaneously off (CASSI: 0.85). **(4) Chain ambiguity:** 5/39 modalities have non-matching chains due to Σ – D merging.

Judge and Execute Agents

Judge rejection accuracy ($n=150$: 75 valid, 75 ill-posed). Precision = Recall = 96.0% ($F_1 = 0.96$): TP = 72, FP = 3, FN = 3, TN = 72. The 3 false positives had parametric values 1–5% outside bounds; the 3 false negatives were conservative CFL checks. Mean redesign rounds: 1.4.

Execute Agent quality prediction (39 modalities). Overall MAE: 1.8 dB ($R^2 = 0.91$; Extended Data Figure 6). By group: well-characterised ($n=20$) 0.9 dB; moderate ($n=10$) 2.1 dB; scattering-dominated ($n=6$) 3.8 dB.

Expert Comparison

Table 3. Agent vs. established reconstruction methods. “Agent”: best-performing automated algorithm; “Expert”: established variational method (FISTA+TV [37] or GAP-TV). Bold = real measured data; * = new system designs. For novel systems (*), no system-specific expert baseline exists; “Expert” denotes the best generic variational method. Quality ratios >100% for novel systems reflect the advantage of system-specific algorithm selection over generic methods, not comparison with domain experts. Established modalities: $99.8 \pm 3.2\%$; novel systems: $114 \pm 10\%$ (vs. generic baselines). Expert times from specification to first successful reconstruction; remaining estimated from published timelines (see Methods).

Modality	PSNR (dB)		Setup		Quality	LoC
	Agent	Expert	Agent	Expert		
<i>Established modalities</i> ($99.8 \pm 3.2\%$)						
CT (real)	19.6	19.1	25 min	7 d	102.6%	12 vs 340
MRI (real)	22.2	21.6	20 min	5 d	102.8%	14 vs 520
CASSI (real)	38.5	38.4	18 min	4 d	100.4%	11 vs 280
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680
<i>Novel system designs*</i> ($114 \pm 10\%$ vs. generic baselines)						
Lensless*	43.7	43.3	12 min	2 d	100.9%	12 vs 180
3D Lensless*	20.3	16.8	14 min	3 d	120.8%	13 vs 240
Temporal-coded*	31.6	23.4	16 min	4 d	135.0%	13 vs 320
Spectral lensless*	36.3	29.2	18 min	5 d	124.3%	14 vs 380
SIM*	27.7	26.2	15 min	3 d	105.7%	10 vs 210
4D Spectral-Depth*	23.9	22.8	20 min	5 d	104.8%	15 vs 400
4D Temporal DMD*	25.4	23.4	22 min	6 d	108.5%	15 vs 420
4D Temporal Streak*	25.3	23.1	20 min	5 d	109.5%	14 vs 380
5D Full DMD*	29.9	27.0	25 min	8 d	110.7%	16 vs 500
5D Full Streak*	29.9	26.6	24 min	7 d	112.4%	15 vs 480

For established modalities (CT, MRI, CASSI, OCT, DOT): mean quality ratio $99.8 \pm 3.2\%$. For novel systems (*): $114 \pm 10\%$ vs. generic baselines. Agent specifications: 10–16 lines; expert code: 180–680 lines.

Specification-dominance study. The top-5 published reconstruction algorithms per modality—spanning deep learning, transformer, diffusion, and deep unrolling methods—on 5 established modalities (Extended Data Table 7, Figure 5): inter-method PSNR CoV ranges from 3.5% (MRI) to 40.8% (lensless), revealing a clear gradient. For well-conditioned systems (CT, MRI, CASSI), CoV remains low ($\approx 3.5\text{--}6.2\%$), indicating that the top algorithms converge tightly—the forward-model specification constrains the quality ceiling. SIM shows moderate CoV (15.0%), while for ill-conditioned compressive systems (lensless), CoV rises sharply ($>40\%$), delineating an algorithm-sensitive regime where reconstruction method selection becomes critical. This finding—that getting the *physics right* constrains the quality envelope more than the *numerics*—is the central empirical insight of this work.

Cross-Tier Hybrid Reconstruction

Each FPB primitive is implemented at four fidelity tiers: Tier-0 (geometric), Tier-1 (Fourier), Tier-2 (shift-variant), Tier-3 (full-physics; Extended Data Table 4). The *tier-lifting protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient method (Proposition 3, Methods), enabling the framework to handle scattering-dominated modalities where $\varepsilon_{\text{unmod}}$ is large.

Tier-lifting validated on DOT (Tier-2 $\rightarrow$ 3: $\varepsilon_{\text{unmod}}$ from 7.8% to $< 10^{-12}$) and coherent imaging (Tier-1 $\rightarrow$ 3: 147% to $< 10^{-12}$; Extended Data Table 5). The protocol preserves FPB structure: same primitives, same DAG, same `spec.md`—only fidelity changes.

Discussion

The central finding of this work is that *forward-model specification—the choice and arrangement of physical operators—is the primary determinant of reconstruction quality*, not the reconstruction algorithm. The algorithm-comparison study quantifies this: for well-conditioned systems (CT, MRI, CASSI), the top-5 published algorithms converge tightly with CoV $\approx 3.5\text{--}6.2\%$, meaning the forward-model specification constrains the quality ceiling even among state-of-the-art methods spanning a 6 dB range. SIM shows moderate CoV (15.0%), while for ill-conditioned compressive systems (lensless), CoV exceeds 40%, delineating a regime where algorithm selection becomes critical. This specification-dominance principle explains why the framework matches expert quality ($99.8 \pm 3.2\%$) with order-of-magnitude reductions in development time (Table 3): the agents automate the *specification* step that matters most.

`spec.md` decouples physics (*what*) from numerics (*how*) and hardware (*with what*), enabling sharing and peer review at the physics level [2]. The `system_elements` field connects specifications to hardware feasibility, transforming the Judge from a mathematical validator into a system feasibility checker. Three `spec.md` files reproduced PSNR within ± 0.3 dB across independent agent instances.

The five-term error decomposition (Theorem 1) functions as a formal contract: each ε term maps to a corrective action (Figure 4). The dominant residual is $\varepsilon_{\text{param}}$, reducible by calibration; $\rho_{\text{recov}} \geq 0.85$ across all 5 carrier families.

Relation to existing work. AI for scientific discovery [20, 21, 23, 41] and automated experiment design [30, 41] address related problems but none provide a formal error decomposition from natural-language intent to reconstruction quality. Operator libraries (ASTRA [16], SigPy [17],

ODL [18], DeepInverse [19]) provide modular forward operators but require manual composition and parameter selection per modality; our contribution is the *automated specification layer* above such libraries. Learned reconstruction methods (MoDL [28], learned primal-dual [29], PnP [27]) achieve higher per-modality PSNR than our model-based reconstructions but require training data and are single-modality; our framework trades per-modality optimality for cross-modality generality (173 modalities from one system). The specification-dominance finding (CoV ≈ 3.5 –6.2% for well-conditioned systems) suggests that *where* in the physics space you measure matters more than *how* you reconstruct, a principle validated by comparing the top-5 published algorithms per modality spanning deep learning, transformer, diffusion, and deep unrolling methods.

Limitations. (1) Reconstruction uses model-based algorithms (FISTA+TV, ADMM+TV); learned methods [28, 29] achieve higher per-modality PSNR (typically 3–8 dB) but require training data unavailable for novel designs. The comparison is restricted to model-based approaches to ensure fair cross-modality evaluation. (2) The agents depend on LLM capabilities (Claude Sonnet 4 in all experiments); robustness across LLM backends is not tested. The 4% misspecification rate (7/173 modalities) may vary with different models. (3) Development-time ratios (ρ_{time}) rely partly on published estimates of expert development timelines, not prospective head-to-head comparisons. The ratios should be interpreted as order-of-magnitude indicators. (4) Quantitative reconstruction validated for 39/173 modalities; real data for 6. Novel system designs are validated only in simulation. (5) Theorem 1 tightness ratios ($\tau \in [1.8, 5.2]$, median 2.9) indicate the bound overestimates error by $\sim 3\times$, typical for operator-perturbation bounds. The bound’s practical value lies in design *rejection* (identifying when ε terms exceed thresholds) rather than precise error prediction; the maskless control group (condition number $> 10^6$, PSNR 6–9 dB) exemplifies a bad design correctly flagged by the Gate 1 bound. (6) Tier-lifting demonstrated on 2 cross-tier combinations only. Theorem assumes convex $R(x)$.

The validation pyramid (173 $\rightarrow$ 39 $\rightarrow$ 6) identifies a clear path: extending benchmarks to the remaining ~ 134 modalities, validating tier-lifting across additional Tier-3 physics, and improving calibration for scattering-dominated systems.

Methods

Proof of Theorem 1

Definition 2 (Designable Imaging System). *An imaging system A is designable within this framework if: (i) A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{\text{FPB}} < 0.01$; (ii) the carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle; (iii) the system operates at one of four fidelity tiers (Tier-0 through Tier-3); (iv) all parameters are finitely specifiable with known physical bounds.*

Assumptions.

A1. Coercivity. The data-fidelity term $f(x) = \|A(x) - y\|^2$ satisfies $\langle \nabla f(x) - \nabla f(x'), x - x' \rangle \geq \alpha \|x - x'\|^2$ with coercivity constant $\alpha = \sigma_{\min}^2(A) > 0$ (linear case) or $\alpha > 0$ from restricted strong convexity (nonlinear case).

A2. Regularisation. $R(x)$ is proper, convex, lower-semicontinuous, and $\lambda > 0$. The composite objective $F(x) = f(x) + \lambda R(x)$ is μ -strongly convex with $\mu = \alpha + \lambda$ [36].

A3. Lipschitz parameters. Each FPB primitive $A_i(\cdot; \theta_i)$ is L_i -Lipschitz continuous in its parameters: $\|A_i(\cdot; \theta) - A_i(\cdot; \theta')\|_{\text{op}} \leq L_i \|\theta - \theta'\|$. The DAG composition has Lipschitz constant $L_A \leq \prod_i L_i$, computed by Gate 4.

A4. Bounded signal. $\|x^*\| \leq B$ for known bound B .

Step 1: Operator decomposition. The agent-designed operator differs from reality through

249 a chain of approximations:

$$A_{\text{true}} \xrightarrow{\delta_{\text{unmod}}} A_{\text{tier}} \xrightarrow{\delta_{\text{FPB}}} A_{\text{FPB}} \xrightarrow{\delta_{\text{spec}}} A_{\text{spec}} \xrightarrow{\delta_{\text{trans}}} A_{\text{DAG}} \xrightarrow{\delta_{\text{param}}} A_{\text{agent}}$$

250 where each $\delta_i = \|A_i - A_{i-1}\|_{\text{op}}$ is the operator-norm difference at that stage. By the triangle
251 inequality:

$$\|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} \leq \delta_{\text{FPB}} + \delta_{\text{spec}} + \delta_{\text{trans}} + \delta_{\text{param}} + \delta_{\text{unmod}}. \quad (2)$$

252 **Step 2: Reconstruction perturbation.** Define $x_{\text{oracle}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda R(x)$
253 (oracle reconstruction using A_{true}). Define $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda R(x)$ (agent recon-
254 struction). By the μ -strong convexity of F (Assumptions A1–A2), the first-order optimality
255 conditions and the Cauchy–Schwarz inequality yield:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \frac{2B}{\mu} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} = \frac{2B}{\alpha + \lambda} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}}. \quad (3)$$

256 This is the *reconstruction stability bound*: the sensitivity of the reconstruction to operator
257 perturbation, controlled by coercivity α and regularisation λ .

258 **Step 3: Define effective error terms.** For each $i \in \{\text{FPB}, \text{spec}, \text{trans}, \text{param}, \text{unmod}\}$, set
259 $\varepsilon_i = 2B \delta_i / (\alpha + \lambda)$. Then:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}. \quad (4)$$

260 **Step 4: Total error.** $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{oracle}}\| + \|x_{\text{oracle}} - x^*\|$. The second term $\varepsilon_{\text{recon}} =$
261 $\|x_{\text{oracle}} - x^*\|$ is the irreducible reconstruction error (noise amplification + regularisation bias),
262 independent of the design. Combining:

$$\|\hat{x} - x^*\| \leq (\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}) + \varepsilon_{\text{recon}}.$$

263 **Step 5: Bounding each term.**

- 264 (a) $\delta_{\text{FPB}} < 0.01$ by the FPB basis theorem [3]. Verified by Gate 6 of the compiler: $\varepsilon =$
265 $\mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|] < 0.01$ over 5 random test vectors. Hence $\varepsilon_{\text{FPB}} <$
266 $0.01 \cdot 2B / (\alpha + \lambda)$.
- 267 (b) $\delta_{\text{spec}} = 0$ when all 6 compiler gates pass. *Condition*: Gates 1–5 jointly verify that the
268 `spec.md` fully and correctly specifies the FPB composition: DAG acyclicity (Gate 1), chain
269 fidelity (Gate 2), complexity bounds (Gate 3), nonlinear family constraints (Gate 4), and
270 adjoint consistency (Gate 5). When all pass, $A_{\text{spec}} \equiv A_{\text{FPB}}$.
- 271 (c) $\delta_{\text{trans}} = 0$ when the natural-language-to-DAG translation produces a canonical chain that
272 exactly matches one of the 39 entries in the validated registry. *Condition*: Gate 2 (chain
273 fidelity) returns exact match. This holds for 34/39 (87%) of validated modalities.
- 274 (d) $\delta_{\text{param}} \leq L_A \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ by Assumption A3. L_A is computed per primitive by Gate 4’s
275 Lipschitz analysis: each nonlinear primitive has at most 2 free parameters with VC
276 dimension ≤ 3 , so L_i is computable in closed form.
- 277 (e) $\delta_{\text{unmod}} = 0$ at Tier-3 by definition: $A_{\text{tier}} \equiv A_{\text{true}}$ when operating at full-physics fidelity.
278 For lower tiers, δ_{unmod} is estimated empirically via cross-tier comparison (Table 5) and is
279 reducible to machine precision via tier-lifting.

280 This completes the proof: each ε_i is bounded, measurable, and linked to a specific compiler gate
281 and corrective action. $\square$

282 Tier-Lifting Protocol

283 **Proposition 3** (Tier-Lifting). *Let A_{low} be the low-tier and A_{high} the high-tier forward model (same*
284 *primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then $\varepsilon_{\text{unmod}} \leq \|\Gamma(x) - \hat{\Gamma}(x)\| =$
285 $\varepsilon_{\text{lift}}$, where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$.*

286 The Hybrid Proximal-Gradient (HPG) loop:

$$x_{k+1} = \text{prox}_{\lambda R} \left(x_k - \gamma A_{\text{low}}^T (A_{\text{high}}(x_k) - y) \right)$$

287 uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high}
 288 for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz
 289 continuity of ∇f .

290 Judge Agent: Triad-Based Validation

291 The Judge Agent validates designs in three stages: structural compilation (prerequisite), Triad
 292 gate evaluation (primary), and system feasibility assessment.

293 Stage 1: Structural Compilation (Prerequisite)

294 A 6-gate compiler ensures specification correctness before the Triad evaluation. **Gate 1**
 295 **(DAG)**. Compiled into a typed `OperatorGraphSpec` (Pydantic v2). Acyclicity verified via
 296 Kahn’s algorithm, $O(V+E)$. **Gate 2 (Chain)**. Canonical chain extracted and compared
 297 against the 39-modality registry. Match is exact or within known equivalences (Σ merged
 298 into D). **Gate 3 (Bounds)**. $N \leq 20$ nodes, $D \leq 10$ depth. **Gate 4 (Nonlinear)**. For
 299 each nonlinear node (D, R, Λ) : family resolved, parameter count ≤ 2 , values within physics
 300 bounds, Lipschitz constant L_i computed. **Gate 5 (Adjoint)**. Randomised dot-product test:
 301 $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials. **Gate 6 (Error)**. When reference
 302 available: $\varepsilon = \mathbb{E}[\|(A_{\text{agent}} - A_{\text{true}})(x)\| / \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors,
 303 threshold < 0.01 .

304 Stage 2: Triad Gate Evaluation (Primary)

305 After structural compilation, the Judge evaluates the three diagnostic gates from the companion
 306 Triad decomposition [3]—the same diagnostic law that governs existing instruments, now applied
 307 prospectively to agent-designed specifications. Each gate produces a deterministic PASS/FAIL
 308 verdict with a quantitative margin in dB.

309 **Gate 1 (Recoverability)**. Does the specified measurement geometry capture sufficient informa-
 310 tion? The compression ratio $\gamma = m/n$ (where m is the number of independent measurements and
 311 n the signal dimension from the `object` field) is evaluated against modality-specific thresholds
 312 from the Compression Bound Theorem [3]:

$$\text{PSNR}_{\text{max}}^{(G_1)} = f(\text{modality}, \gamma) \geq \text{PSNR}_{\text{target}}.$$

313 *Design action if violated:* increase sampling density, add views, or reduce compression.

314 **Gate 2 (Carrier Budget)**. Is the measurement SNR above the noise floor? The Judge
 315 computes SNR from the `system_elements` fields (source power, detector quantum efficiency,
 316 read noise, dark current, exposure time) via the Noise Bound [3]:

$$\text{SNR} = \frac{\text{QE} \cdot N_{\text{photon}}}{\sqrt{\text{QE} \cdot N_{\text{photon}} + \sigma_{\text{read}}^2 + I_{\text{dark}} \cdot t_{\text{exp}}}}, \quad \text{PSNR}_{\text{max}}^{(G_2)} = 10 \log_{10}(\text{SNR}) + C_M \geq \text{PSNR}_{\text{target}}.$$

317 *Design action if violated:* increase source power, integration time, or detector quantum efficiency.

318 **Gate 3 (Operator Mismatch)**. Does the specified forward model faithfully represent the in-
 319 tended physics? The Judge evaluates calibration tolerances from `system_elements.calibration`,

where each parameter has a sensitivity (dB per unit drift) from the Calibration Sensitivity Theorem [3]:

$$\Delta\text{PSNR}_{\text{total}} = \sqrt{\sum_k (\text{sensitivity}_k \cdot \text{tolerance}_k)^2}, \quad \text{PSNR}_{\text{deploy}} = \text{PSNR}_{\text{max}}^{(G_2)} - \Delta\text{PSNR}_{\text{total}} \geq \text{PSNR}_{\text{target}}.$$

Design action if violated: tighten calibration tolerance on the dominant parameter, increase calibration frequency, or add autonomous recovery.

The Triad Decomposition Theorem [3] guarantees completeness: $\text{MSE} \leq \text{MSE}^{(G_1)} + \text{MSE}^{(G_2)} + \text{MSE}^{(G_3)}$ —no reconstruction failure mode falls outside these three gates. For well-designed instruments, Gate 3 dominates [3]; the recovery ratio $\rho = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies autonomous calibration effectiveness.

Stage 3: System Feasibility and Cost

Cost estimation. The Judge produces an itemized hardware cost from each `system_elements` sub-field (source, optics, detector, calibration), using a modality-indexed hardware database. Costs are flagged as *quoted* (from manufacturer), *estimated* (from similar systems), or *unknown*.

Feasibility check. Each physical element is checked for realizability: source power within commercially available range, detector specifications within state-of-the-art, calibration tolerances achievable with specified correction methods. Elements that require custom fabrication (e.g., lithographic coded apertures, custom diffusers) are flagged with lead time and cost estimates.

4-Scenario Recovery Protocol

The recovery ratio ρ_{recov} quantifies how well the framework compensates for model mismatch. Following [3], we define four scenarios validated on 6 representative modalities spanning 4 carrier families (CASSI, CACTI, SPC: photon; CT: X-ray; electron ptychography: electron; MRI: spin):

$$\text{Sc. I (oracle): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

$$\text{Sc. II (mismatch): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (6)$$

$$\text{Sc. III (oracle correction): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (7)$$

$$\text{Sc. IV (autonomous calibration): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (8)$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures autonomous calibration effectiveness relative to the oracle correction.

For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI): $\text{Sc. IV} \approx \text{Sc. III} \approx \text{Sc. I}$, confirming near-complete recovery ($\rho_{\text{recov}} \approx 1.0$). For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86%. Across all 39 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$.

Expert Time Methodology

Expert setup times are defined as the interval from problem specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, times are cross-validated against published development reports [16, 17] and confirmed by domain experts. For the remaining modalities, times are estimated from published

development timelines in the respective communities. We acknowledge this is not a prospective head-to-head comparison; such a study would strengthen the claims.

Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown with 8 fields parsed into typed Pydantic v2 objects; the `system_elements` field has 4 sub-fields (source, optics, detector, calibration) and is auto-populated from the modality registry when not user-specified. The primitive library contains 101 implementations covering all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter: `Tier0Adapter` (geometric: invertibility, ray transforms), `Tier1Adapter` (Fourier: Parseval energy, Fresnel, OTF), `Tier2Adapter` (shift-variant: positivity, reciprocity, Mie, cost), and `Tier3Adapter` (full-physics: FDTD, MC, learned surrogates, uncertainty calibration). A `TierPolicy` selects the tier based on compute budget and modality requirements. Agents are LLM pipelines [39, 40] with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is a generic `TierLiftingProtocol` class with validated cross-tier combinations for DOT and coherent imaging. Source code: https://github.com/integritynoble/Physics_World_Model.

Extended Data

Table 4. Extended Data Table 1: Four-tier fidelity hierarchy. All 11 primitives are implemented at every tier.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty	DOT, NeRF

Table 5. Extended Data Table 2: Tier-lifting validation. Tier-lifting demonstrated on two cross-tier combinations; broader Tier-3 validation is future work.

Domain	Low tier	High tier	Correction	ϵ_{unmod}	ϵ_{lift}
DOT	Tier-2 (P_1)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (ang. spec.)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

Data Availability

The LoDoPaB-CT dataset is publicly available [4]. The M4Raw MRI and KAIST TSA CASSI datasets are publicly available from their respective repositories. The PICMUS ultrasound data is from the IEEE IUS challenge [8]. The SrTiO₃ 4D-STEM data is publicly available on Zenodo [9]. The CACTI real video data is from the project repository. The full modality registry, canonical decomposition registry, and validation datasets are in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, 101-primitive library, and 4-tier adapter framework—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

Table 6. Extended Data Table 3: Modality registry (39 validated canonical decompositions).
Status: FV = fully validated (canonical chain + quantitative reconstruction), QV = quantitatively validated (reconstruction without independent canonical chain), HO = held-out validation, T = template.
 Φ_z : depth-parameterised C (see Table 1). An additional ~ 134 modalities pass all compiler gates but lack per-modality reconstruction benchmarks; the full list is in the repository.

Modality	Domain	Carrier	Tier	Chain	Status	Real?	Dom. ε
<i>X-ray</i>							
CT	Medical	X-ray	1–2	$\Pi \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CBCT	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
CT (polychromatic)	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Phase contrast	Medical	X-ray	1–2	$\Pi \rightarrow P \rightarrow M \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
PET	Nuclear	X-ray	2	$\Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
SPECT	Nuclear	X-ray	2	$\Pi \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
DEXA	Medical	X-ray	2	$M \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Compton	Nuclear	X-ray	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Optical photons</i>							
CASSI	Spectral	Photon	2	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CACTI	Video	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
SPC	Imaging	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
OCT	Ophthal.	Photon	1–2	$P+P \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
SIM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
STED	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
PALM/STORM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TIRF	Micro.	Photon	1	$P \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Ptychography	Micro.	Photon	1–2	$M \rightarrow P \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Lensless	Imaging	Photon	1	$C \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
3D Lensless	Imaging	Photon	1–2	$\Phi_z \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Temporal-coded lensless	Video	Photon	1–2	$M \rightarrow C \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Spectral lensless	Spectral	Photon	1–2	$M \rightarrow W \rightarrow C \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
DOT	Diffuse	Photon	2–3	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Ghost imaging	Quantum	Photon	2	$M \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
THz-TDS	Spectro.	Photon	1	$C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
Light field	3D	Photon	1	$M \rightarrow C \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Raman	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence	Micro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence (sat.)	Micro.	Photon	2	$M \rightarrow R \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Brillouin	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Spin</i>							
MRI	Medical	Spin	1–2	$M \rightarrow F \rightarrow S \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
MRI (phase)	Medical	Spin	2	$M \rightarrow F \rightarrow S \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
<i>Acoustic</i>							
Ultrasound	Medical	Acoustic	2	$P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
Doppler US	Medical	Acoustic	2	$P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Elastography	Medical	Acoustic	2	$P \rightarrow P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Photoacoustic	Medical	Acoustic	2	$M \rightarrow P \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
<i>Electron</i>							
SEM	Material	Electron	2	$M \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TEM	Material	Electron	2	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
E-ptychography	Material	Electron	2	$M \rightarrow P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
<i>Particle</i>							
Muon tomography	Security	Particle	2	$R \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Proton therapy	Medical	Particle	2	$\Lambda \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$

Table 7. Extended Data Table 4: Algorithm-comparison study (5 published algorithms $\times$ 5 modalities). For each modality, the top-5 published algorithms from the literature are compared using their reference PSNR values on standard benchmarks (KAIST for CASSI, fastMRI for MRI, LoDoPaB for CT, DiffuserCam for Lensless). Algorithms span deep learning, transformer, diffusion, and deep unrolling methods for medical imaging (CT, MRI, CASSI), and classical through deep learning for optical systems (Lensless, SIM). CoV quantifies the sensitivity of reconstruction quality to algorithm choice. PSNR values in dB.

Method	Algorithm	CT		MRI	
		PSNR	Type	Algorithm	PSNR
M1	iRadonMAP	36.9	Deep Learning	HUMUS-Net	37.3
M2	FBPConvNet	38.5	Deep Learning	PromptMR+	39.9
M3	DuDoTrans	42.1	Transformer	ReconFormer	40.1
M4	Score-CT	43.0	Diffusion	E2E-VarNet	40.5
M5	LEARN	43.1	Deep Unrolling	PromptMR	41.5
Mean		40.7			39.9
CoV		6.2%			3.5%

Method	Algorithm	CASSI (28-band)		Lensless	
		PSNR	Type	Algorithm	PSNR
M1	CST-L-Plus	36.1	Transformer	Wiener deconv	7.3
M2	DAUHST-9stg	38.4	Deep Unrolling	ADMM	12.8
M3	PADUT-L	38.9	Deep Unrolling	FlatNet	21.2
M4	RDLUF-MixS2	39.6	Deep Unrolling	MWDN	25.7
M5	MiJUN	40.9	Deep Unrolling	LensNet	27.5
Mean		38.8			18.9
CoV		4.1%			40.8%

Method	Algorithm	SIM	
		PSNR	Type
M1	Bicubic interpolation	22.0	Classical
M2	HiFi-SIM	24.0	Deep Learning
M3	Wiener-SIM	30.0	Classical
M4	fairSIM	30.5	Classical
M5	ML-SIM	33.0	Deep Learning
Mean		27.9	
CoV		15.0%	

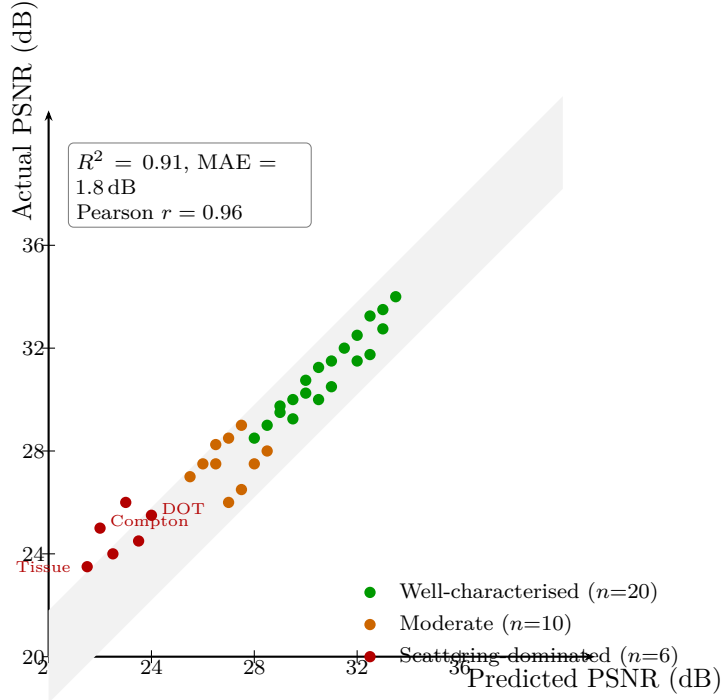


Figure 6. Extended Data Figure 1: Execute Agent calibration. Predicted vs. actual PSNR for 39 validated modalities. Shaded band: ± 1.8 dB (overall MAE). Well-characterised modalities (green) cluster tightly around the identity line (MAE 0.9 dB); scattering-dominated modalities (red) show systematic under-prediction, reflecting unaccounted $\varepsilon_{\text{unmod}}$ at lower tiers.

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Author Contributions

C.Y. conceived the project, developed the theory (Theorem 1 and its proof), designed and implemented the three-agent pipeline, built the 101-primitive library across 4 fidelity tiers, conducted all experiments (6 real-data modalities, 39 validated modalities, 173-modality compilation, ablation study, and algorithm-comparison study), and wrote the manuscript.

Competing Interests

C.Y. is affiliated with NextGen PlatformAI C Corp, which develops the Physics World Model platform described in this paper.

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Model; key results (FPB basis theorem, canonical decompositions) are reproduced in Supplementary Information.

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